1증명과제

1. $\vec{A}: A_x\hat{a_x}+A_y\hat{a_y}A_z\hat{a_z}$

$\vec{B}: B_x\hat{a_x}+B_y\hat{a_y}B_z\hat{a_z}$

$\vec{C}: C_x\hat{a_x}+C_y\hat{a_y}C_z\hat{a_z}$

$\vec{A}\cdot(\vec{B}\times\vec{C})=\vec{B}\cdot(\vec{A}\times\vec{C})=\vec{C}\cdot(\vec{A}\times\vec{B})$

$\vec{A}\cdot(\vec{B}\times\vec{C})$

$=\vec{A}\cdot(B_xC_x\hat{a_x^2}+B_xC_y\hat{a_x}\hat{a_y}+B_xC_z\hat{a_x}\hat{a_z}+B_yC_x\hat{a_y}\hat{a_x}+B_yC_y\hat{a_y}\hat{a_y}+B_yC_z\hat{a_y}\hat{a_z}+B_zC_x\hat{a_z}\hat{a_x}+B_zC_y\hat{a_z}\hat{a_y}+B_zC_z\hat{a_z^2})$

$=\vec{A}\cdot(-B_xC_y\hat{a_z}+B_xC_z\hat{a_y}+B_yC_x\hat{a_z}-B_yC_z\hat{a_x}-B_zC_x\hat{a_y}+B_zC_y\hat{a_x})$

$=(A_x\hat{a_x}+A_y\hat{a_y}+A_z\hat{a_z})\cdot(-B_xC_y\hat{a_z}+B_xC_z\hat{a_y}+B_yC_x\hat{a_z}-B_yC_z\hat{a_x}-B_zC_x\hat{a_y}+B_zC_y\hat{a_x})$

$=(A_x\hat{a_x}+A_y\hat{a_y}+A_z\hat{a_z})\cdot\hat{a_x}(B_zC_y-B_yC_z)+\hat{a_y}(B_xC_z-B_zC_x)+\hat{a_z}(B_yC_x-B_xC_y)$

$=\hat{a_x}(A_xB_zC_y-A_xB_yC_z)+\hat{a_y}(A_yB_xC_z-A_yB_zC_x)+\hat{a_z}(A_zB_yC_x-A_zB_xC_y)$

$\vec{B}\cdot(\vec{A}\times\vec{C})$

$=\vec{B}\cdot(-A_xC_y\hat{a_z}+A_xC_z\hat{a_y}+A_yC_x\hat{a_z}-A_yC_z\hat{a_x}-A_zC_x\hat{a_y}+A_zC_y\hat{a_x})$

$=\hat{a_x}(B_xA_zC_y-B_xA_yC_z)+\hat{a_y}$

2.증명과제

$\vec{A}:(A_x,A_y,A_z),\vec{B}:(B_x,B_y,B_z),\vec{C}:(C_x,C_y,C_z)$

$\vec{B}\times\vec{C}=(B_yC_z-B_zC_y,B_zC_x-B_xC_z,B_yC_x-B_xC_y)$이다

$\vec{A}\times(\vec{B}\times\vec{C})=$

$[A_y(B_xC_y-B_yC_x)-A_z(B_zC_x-B_xC_z),A_z(B_yC_z-B_zC_y)-A_x(B_xC_y-B_yC_x),A_x(B_zC_x-B_xC_z)-A_y(B_yC_z-B_zC_y)]$

전개하면

$(A_yB_xC_y-A_yB_yC_x-A_zB_zC_x+A_zB_xC_z,A_zB_yC_z-A_zB_zC_y-A_xB_xC_y+A_xB_yC_x,A_xB_zC_x-A_xB_xC_z-A_yB_yC_z+A_yB_zC_y)$이다

각각B,C에 정리하면

$(B_x(A_yC_y+A_zC_z)-C_x(A_yB_y+A_zB_z),B_y(A_xC_x+A_zC_z)-C_y(A_xB_x+A_zB_z),B_z(A_xC_x+A_yC_y)-C_z(A_xB_x+A_yB_y))$

공토요소를 묶으면

$B_x(A_yC_y+A_zC_z)+B_y(A_xC_z+A_xC_z)+B_z(A_xC_x+A_yC_y)-C_x(A_yB_y+A_zB_z)-C_y(A_zB_z+A_xB_x)-C_z(A_xB_x+A_yB_y)$

여기에 $A_xB_xC_x,A_yB_yC_y,A_zB_zC_z$를 각각 더해주고 빼주면

$(A\cdot C)B_x+(A\cdot C)B_y+(A\cdot C)B_z-(A\cdot B)C_x-(A\cdot B)C_y-(A\cdot B)C_z$

이므로 $(A\cdot C)B-(A\cdot B)C$

3.증명과제

$\int_{S_1}\vec{E_1}d\vec{A}+\int\vec{E_2}d\vec{A}=0$ 이므로 $E\int_{S_1}dA+E\int_{S_2}dA=0$

(-x방향) (+x방향)

-x방향: $E\int_{S_1}dA=E\times\pi r=\frac{\lambda 2r}{\epsilon_0^2}$---> 따라서 $E=\frac{\lambda 2r}{\epsilon_0\pi r^2}\frac{\hat{l}}{2}$

+x방향:$E\int_{S_2}dA=E\times\pi r^2=\frac{\lambda 2r}{\epsilon_0}$---> 따라서 $E=\frac{\lambda 2r}{\epsilon_0\pi r^2}\frac{\hat{l}}{2}$

-x방향과 +x방향의 E값이 같으므로 서로 상쇄되어 S_1과S_2의 기여도가 0이된다.

1.2번 점$P_1(x_1,y_1,z_1)$와$P_2(x_2,y_2,z_2)$로향하는선분벡터의단위벡터를
이용하여직각좌표계를표현하여라
$(x_2-x_1)\vec{A_x}+(y_2-y_1)\vec{A_y}+(z_2-z_1)\vec{A_z}$
1.7번 벡터$A=A_xa_x+2a_y+A_za_z, B=2a_x+3a_y+4a_z$가평행하기위한조건은?
평행하기 위해선 계수의 비가 같아야 하므로
$A=A_xa_x+2a_y+A_za_z$
$B=2a_x+3a_y+4a_z$
A에$\frac{3}{2}$를하면B와 같아지므로
$\frac{3}{2}A_xa_x+3a_y+\frac{3}{2}A_za_z=2a_x+3a_y+4a_z$이므로
$A_x=\frac{4}{3}, A_z=\frac{8}{3}$이다

1.8번 벡터$A=a_x-a_y-a_z$와$B=a_x-2a_z$가있다이때크기는6이고
A,B가평행하기위한벡터C는?
1, $\sqrt[2]{a^2+b^2+c^2}=6$
2, $a-b-c=0$
3, $a-2b=0$
1,2,3을연립하면$a=2b, b=c, b=\pm\sqrt[2]{6}$이나온다
따라서$C=(\pm2\sqrt[2]{6},\pm\sqrt[2]{6},\pm\sqrt[2]{6})$이다
1.10번 $B=(3x^2+2y)a_x+xya_y+2xa_z$일 때 점$P(1,-2,2)$에서
벡터B에 평행한단위벡터는?
평행한벡터를R이라할때
$R=-a_x+(-2)a_y+2a_z$
$\hat{R}=\frac{R}{|R|}=\frac{-a_x-2a_y+2a_z}{\sqrt[2]{9}}=-\frac{1}{3}a_x-\frac{2}{3}a_y+\frac{2}{3}a_z$이다

1.13번 $A=4a_x-3a_y-2a_z, B=2a_x-2a_y-a_z$는평행사변형의
두벡터를표시하는벡터이다이평행사변형의크기는?
$|A\times B|=a_x\begin{vmatrix}-3&-2\\-2&-2\end{vmatrix}-a_y\begin{vmatrix}4&-2\\2&-1\end{vmatrix}+a_z\begin{vmatrix}4&-3\\2&-2\end{vmatrix}=|-a_x-0a_y-2a_z|$
$=|-a_x-2a_z|=\sqrt[2]{5}$이다
1.16번 $A\cdot(A\times B)$를증명하여라
x,y,z공간에A를$+x$방향이라하고B를$+y$방향이라하면
$A\times B$가$+z$방향이고
$A\times B$가A와직교하므로0이다

1.19번 z축상한점h에서$z=0$인원통표면의한점$P(\rho,\phi,0)$으로
향하는선분벡터의단의벡터는?
선분벡터$\vec{R}:(\rho,\phi,-z)$
$\hat{R}=\frac{R}{|R|}=\frac{1}{\sqrt[2]{\rho^2+\phi^2+z^2}}(\rho,\phi,z)=\frac{\rho}{\sqrt[2]{\rho^2+\phi^2+z^2}}+\frac{\phi}{\sqrt[2]{\rho^2+\phi^2+z^2}}-\frac{z}{\sqrt[2]{\rho^2+\phi^2+z^2}}$
1.21번 $R=\rho\sin a_\rho+\rho\cos a_\phi+za_z$의벡터계가있다
점$P(3,\frac{\Pi}{2},4)$에서원통측면에수직한성분벡터와평행한성분벡터는?

1.25번 직각좌표계로표현된벡터$B=ya_x-xa_y+za_z$를원통좌표계로변환하라
원통좌표계로변환된벡터B가ρ,ϕ,z방향으로각각B_ρ,B_ϕ,B_z의성분을갖는다면
$B=B_\rho a_x+B_\phi a_y+B_z a_z$이다
$x=\rho\cos\phi, y=\rho\sin\phi$
$B_\rho=B\cdot a_\rho=(ya_x-xa_y+za_z)\cdot a_\rho=ya_xa_\rho-xa_ya_\rho+za_za_\rho=ycos\phi-xsin\phi$
$=\rho\sin\phi\cos\phi-\rho\sin\phi\cos\phi=0$
$B_\phi=B\cdot a_\phi(ya_x-xa_y+za_z)\cdot a=ya_xa_\phi-xa_ya_\phi+za_za_\phi=-ysin\phi-xcos\Phi$
$=-\sin^2\phi-\cos^2\phi=-\rho$
$B_z=z\rightarrow B_z=-\rho a_\phi+za_z$

1.26번 구좌표계로표현된벡터$B=\frac{1}{r}a_r$을직각좌표계로변환하라
$x=rsin\theta\cos\phi, y=rsin\theta\sin\phi, z=rcos\theta$
$\rightarrow x^2+y^2=r^2\sin^2\theta(\sin^2\phi+\cos^2\phi)=r^2\sin^2\theta$
$\rightarrow\sin\theta=\frac{\sqrt[2]{x^2+y^2}}{r},\cos\theta=\frac{z}{r}\rightarrow\sin\phi=\frac{y}{rsin\theta}=\frac{y}{\sqrt[2]{x^2+y^2}},\cos\phi=\frac{x}{rsin\theta}=\frac{x}{\sqrt[2]{x^2+y^2}}$
$B_x=B\cdot a_x=\frac{1}{r}\sin\theta\cos\phi=\frac{x}{r^2},B_y=B\cdot a_y=\frac{1}{r}\sin\theta\sin\phi=\frac{y}{r^2},B_z=B\cdot a_z=\frac{1}{r}\cos\theta=\frac{z}{r^2}$
$B=\frac{1}{r^2}(xa_x+ya_y+za_z)$

연습문제2.3
$P1$에서받는전계E
$E=\frac{Q}{4\pi\epsilon_0 r^2}\hat{a}=\frac{3\times10^{-9}}{4\pi\epsilon_0\times4}\cdot\frac{2}{2}\hat{a_z}=\frac{1}{4\pi\epsilon_0}\times\frac{3\cdot10^{-9}}{4}\hat{a_z}=\frac{27}{4}\hat{a_z}=6.75\hat{a_z}$
$P2$에서받는전계E
$E=\frac{Q}{4\pi\epsilon_0 r\cdot2}\hat{R}=\frac{3\times10^{-9}}{4\pi\epsilon_0\times20}\times(\frac{4}{\sqrt[2]{20}}a_x+\frac{2}{\sqrt[2]{20}}a_z)=\frac{1}{4\pi\epsilon_0}\cdot\frac{3\times10^{-9}}{5\sqrt[2]{20}}a_x+\frac{1}{4\pi\epsilon_0}\cdot\frac{3\times10^{-9}}{16\sqrt[2]{20}}a_z=1.2\hat{a_x}+0.6\hat{a_z}$
$P2+P1=1.2\hat{a_x}+7.35\hat{a_z}$

연습문제2.6
xy평면상이고$\parallel(0,0,4)$의전계$\rightarrow$전원점$(0,0,0)$
$a_R=\frac{4}{4}\hat{a_z}=\hat{a_z}$
$E=\frac{\rho_L}{2\pi\epsilon_0 R}a_R=\frac{2\pi\epsilon_0}{2\pi\epsilon_0\cdot4}a_z=\frac{1}{4}a_z$이므로크기는$\frac{1}{4}$이다

연습문제2.14
$Q=\int_s P_s ds\rightarrow\Psi=\int P_s ds=\int_0^{2\pi}\int_0^4 2P\cdot PdPd\Phi=\int_0^{2\pi}\int_0^4 2P^2dPd\Phi=[\frac{2}{3}P^3]_0^4\cdot2\pi=\frac{256}{3}\pi$

연습문제2.21
$D=x^3y^3a_x+3x^2yz^2a_y+4xyza_z$
직각좌표계$\rightarrow\div D=\frac{\sigma D_x}{\sigma x}+\frac{\sigma D_y}{\sigma y}+\frac{\sigma D_z}{\sigma z}=3x^2y^3+3x^2z^2+4xy$
$P(1,2,10$에서의체적전하밀도,즉전곳밀도의발산도$\div D_p=24+3+8=35$이다

연습문제2.35
$z=0\ \ 0<P<4\ \ \rho_s=10\times10^{-9}[C/m^2]$ z축상$P(0,0,5)$
$ds=\rho d\rho d\Phi$
$R=2a_z-\rho a_p=5a_z-\rho a_p\rightarrow V=\int_s\frac{\rho_s}{4\pi\epsilon_0 R}ds=\int_0^{2\pi}\int_0^4\frac{10\times10^{-9}}{4\pi\epsilon_0\sqrt{25+\rho^2}}\rho d\rho d\Phi\rightarrow$치환. $\sqrt{25+\rho^2}=t\rightarrow t^2=25+\rho^2$
$\rightarrow$미분 $2tdt=2\rho d\rho\rightarrow tdt=\rho d\rho$
$dt=\frac{\rho}{t}d\rho=\frac{\rho}{\sqrt{25+\rho^2}}d\rho$대입$\rightarrow V=\int_0^{2\pi}\int_0^4\frac{10\times10^{-19}}{4\pi\epsilon_0}\times\frac{1}{\sqrt{25\times\rho^2}}\rho d\rho d\Phi=\int_0^{2\pi}\int_{\sqrt{100}}^{\sqrt{116}}\frac{10\times10^{-19}}{4\pi\epsilon_0}\times\frac{1}{\sqrt{25\times\rho^2}}tdtd\Phi$
$=\int_0^{2\pi}\int_{\sqrt{100}}^{\sqrt{116}}\frac{10\times10^{-19}}{4\pi\epsilon_0}\times d\Phi dt=9\times10^{-9}\times10\times10^{-9}\times2\pi\int_{10}^{\sqrt{116}}dt=90\times4.84\fallingdotseq435.611$